
Application of the Hopfield network associative memory for the reconstruction and forecasting of stock market time series

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Abstract

Associative memory offers a natural framework for testing whether recurring local structures in financial time series can be stored, retrieved, and reused for inference. This study evaluates a classical binary Hopfield network on daily financial data for SPY, AAPL, and TSLA from 2020 to 2025. Logarithmic returns are transformed into bipolar sign patterns, with additional experiments using a binary volatility channel. We first validate the implementation through synthetic tests and stored-pattern recall under different memory loads and corruption levels. We then move to the harder setting of previously unseen temporal windows, where the network must reconstruct artificially corrupted test patterns using only historical memories. Although stored patterns are recalled reliably at low memory load, unseen-pattern reconstruction does not provide systematic improvement: Hopfield dynamics tends to move test patterns toward memory-supported attractors without improving their similarity to the original clean test state. Prototype-based memory selection improves historical coverage in some settings but does not remove this limitation. For one-step direction forecasting, classical Hopfield retrieval is compared with majority, persistence, first-order Markov, nearest-analogue, k-nearest-neighbour, and logistic-regression baselines. No method shows a stable cross-asset advantage, and classical Hopfield dynamics does not systematically improve direct historical retrieval. An asymmetric temporal associative model produces isolated gains for SPY and TSLA at selected window lengths, but the effect is not universal. The results identify a coverage–interference trade-off as a central limitation of classical Hopfield memory for structured financial sequences and show that temporal scale affects performance without yielding a single consistently superior scale.

Keywords: Hopfield network; associative memory; financial time series; stock returns; pattern reconstruction; directional forecasting; temporal scale; Hurst exponent.

Introduction

Financial time series are difficult to model because their statistical structure is noisy, nonstationary, and regime-dependent. Empirical research has documented volatility clustering, heavy tails, scaling behaviour, and other departures from simple independent Gaussian dynamics [7, 8]. At the same time, evidence for long-range dependence in raw stock returns is not universal; for example, Lo showed that apparent long-memory effects can disappear after short-range dependence is controlled for [6]. This motivates a cautious view of temporal regularity: local structures may recur, but recurrence does not automatically imply stable predictability.

Machine-learning methods have been applied extensively to financial forecasting, from classical classifiers to deep recurrent and attention-based architectures [4, 9, 11]. Reported improvements depend strongly on the target variable, feature set, horizon, validation design, and market regime. Directional prediction of the next return is especially demanding because the available signal is weak relative to short-horizon noise, and performance claims are highly sensitive to leakage and baseline choice [4].

Hopfield networks provide a different perspective. Instead of learning a conventional input–output mapping, the classical Hopfield model stores patterns as attractors of a recurrent dynamical system [2]. A corrupted state can then evolve toward a nearby stored memory. For random, weakly correlated patterns, classical analyses show a limited storage regime and a sharp degradation of retrieval as the memory load grows [1]. This creates an immediate question for financial data: if local temporal patterns repeat, can associative memory recover noisy market fragments or identify historical analogues useful for forecasting?

Recent work has substantially expanded the Hopfield framework. Dense and modern Hopfield networks offer much higher memory capacity and connections to attention mechanisms [5, 10]. Modern Hopfield layers have also been incorporated directly into time-series prediction architectures, including the multi-resolution STanHop model [13]. In a different direction, structured-memory studies have shown that Hopfield-like systems can exhibit generalization to previously unseen examples when training patterns are generated by shared latent features [3]. These results motivate investigating whether correlated financial windows exhibit any analogous associative structure, while also making clear that such behaviour is not guaranteed for an arbitrary real-world dataset.

This work focuses deliberately on the classical binary Hopfield model rather than a modern deep architecture. The aim is to isolate what the original associative-memory mechanism can and cannot extract from financial temporal patterns. The study addresses four questions: (i) whether financial sign patterns stored in memory can be robustly recalled under corruption; (ii) whether previously unseen financial fragments can be reconstructed through association with historical memories; (iii) whether associative retrieval provides useful information for one-step return-direction forecasting; and (iv) whether the outcome depends systematically on temporal window length. To separate implementation failure from model limitations, the experiments progress from stored-pattern controls to unseen reconstruction, memory-selection variants, forecasting baselines, and finally an asymmetric temporal associative model inspired by sequence-association formulations [12]. The central hypothesis is that recurrent local structure in financial time series can be exploited by associative memory: a Hopfield network trained on historical temporal patterns should be able to improve corrupted previously unseen fragments by association with stored states and, if retrieved historical analogues contain continuation information, provide above-baseline information about the sign of the next return. Because financial dynamics may depend on temporal scale, both effects are expected to vary with the window length L .

Materials and Methods

Financial data and preprocessing

Daily adjusted closing prices for SPY, AAPL, and TSLA were downloaded using 1 January 2020 as the start date and 31 December 2025 as the exclusive end date. Missing observations were forward-filled before return calculation. Logarithmic returns were computed as

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right), \quad (1)$$

where P_t denotes the adjusted closing price at time t . 57

For the classical Hopfield network, returns were converted to bipolar states 58

$$x_t = \begin{cases} +1, & r_t \geq 0, \\ -1, & r_t < 0. \end{cases} \quad (2)$$

A temporal pattern of length L is therefore a vector $X_t = [x_{t-L+1}, \dots, x_t] \in \{-1, +1\}^L$. 59
The investigated window lengths were 60

$$L \in \{10, 30, 50, 70, 100, 200\}. \quad (3)$$

A second binary representation was constructed to retain information about 61
movement magnitude. For each training split, the volatility threshold was defined as the 62
median absolute training return. Values above this threshold were encoded as $+1$ and 63
the remaining values as -1 . The combined representation concatenated the sign and 64
volatility channels, resulting in a $2L$ -dimensional bipolar vector. Thresholds were 65
estimated from training data only. 66

Yearly descriptive statistics included mean return, standard deviation, 67
positive-return ratio, skewness, kurtosis, and the Hurst exponent. The Hurst statistic 68
was used only as a descriptive indicator of scale-dependent behaviour, not as direct 69
evidence of predictability. This distinction is important because both classical 70
long-memory tests and multifractal models show that different notions of scaling and 71
dependence need not imply forecastability of signed returns [6, 7]. 72

Classical Hopfield network 73

The classical network consists of N bipolar neurons $s_i \in \{-1, +1\}$. Given P stored 74
patterns ξ^μ , the weight matrix is constructed using a Hebbian rule 75

$$w_{ij} = \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu, \quad i \neq j, \quad (4)$$

with $w_{ii} = 0$. The conventional factor $1/N$ is omitted because, under zero thresholds, 76
multiplication of all weights by the same positive constant does not change the sign of 77
the local field and therefore does not change the binary update trajectory. 78

Neurons are updated asynchronously in a random order. The local field for neuron i 79
is 80

$$h_i = \sum_j w_{ij} s_j, \quad (5)$$

and the neuron is set to $+1$ for $h_i > 0$, to -1 for $h_i < 0$, and left unchanged when 81
 $h_i = 0$. With a symmetric zero-diagonal weight matrix, the energy 82

$$E(\mathbf{s}) = -\frac{1}{2} \mathbf{s}^T W \mathbf{s} \quad (6)$$

is non-increasing under asynchronous deterministic updates [2]. A run is considered 83
converged when a full sweep over all neurons leaves the state unchanged. 84

The memory load is defined as 85

$$\alpha = \frac{P}{N}. \quad (7)$$

For the stored-pattern recall experiment, $\alpha \in \{0.10, 0.20, 0.30, 0.50\}$. For unseen-pattern 86
experiments, lower loads $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$ were used to emphasize regimes in 87
which the network itself remains capable of reliable memory retrieval. The classical 88
asymptotic capacity value near $\alpha \approx 0.14$ applies to random uncorrelated patterns and is 89
therefore treated as a reference rather than a universal threshold for strongly correlated 90
sliding financial windows [1]. 91

Implementation validation and stored-pattern recall

Before testing financial generalization, the implementation was validated on small synthetic bipolar patterns. The checks confirmed a symmetric zero-diagonal weight matrix, successful correction of a one-bit corruption, and non-increasing energy. A stored-pattern recall experiment was then performed on financial windows. Stored memories were corrupted by flipping a prescribed fraction of bits and were passed through Hopfield dynamics. The requested corruption levels ranged from 5% to 40%. Because short windows permit only discrete flip counts, analysis used the actual realized corruption fraction whenever necessary.

Stored-pattern recall serves as a control experiment: it verifies whether the implementation can perform the task for which the classical Hopfield model was designed. It is not used as evidence of generalization to previously unseen financial states.

Unseen-pattern reconstruction

For the principal reconstruction experiment, each financial series was divided chronologically into an 80% training part and a 20% test part before sliding windows were constructed. This prevents train/test leakage through overlapping windows across the temporal split. Hopfield memories were formed only from training windows, whereas clean test windows were treated as unseen patterns.

A clean unseen pattern X was corrupted by flipping a subset of its bits, producing X_{noisy} . Hopfield dynamics returned X_{restored} . Reconstruction quality was measured relative to the original clean test pattern. The principal measure was

$$\Delta A = A(X_{\text{restored}}, X) - A(X_{\text{noisy}}, X), \quad (8)$$

where $A(\cdot, \cdot)$ denotes bitwise accuracy. Positive ΔA indicates net correction, whereas negative ΔA indicates that the network introduced more errors than it removed. Hamming distance, exact recovery, convergence, and the number of sweeps were also recorded.

Two memory-selection strategies were compared. Uniform selection sampled historical windows approximately evenly across the training period. Prototype selection used k -means clustering and selected the actual historical pattern closest to each cluster centre. Because bipolar vectors are used, squared Euclidean distance is proportional to Hamming distance. Prototype selection therefore changes which historical states are retained without introducing non-binary artificial memories.

For the main noisy-unseen experiment, 20 test patterns were selected approximately uniformly across each test interval for every asset and window length. Three independent corruptions were generated for each selected pattern and corruption level. Paired corruptions were used when comparing uniform and prototype memories, so that both memory-selection strategies received exactly the same damaged test patterns.

Historical-analogue forecasting

The forecasting target was the sign of the next daily log return. For a context

$$X_t = [x_{t-L+1}, \dots, x_t], \quad (9)$$

the target is $y_{t+1} = x_{t+1}$. Historical training contexts were paired with their observed continuations. The same chronological 80/20 split used in unseen reconstruction was retained, and no context crossing the split boundary was used.

The following forecasting baselines were evaluated: majority class, persistence of the latest sign, a first-order Markov rule, and nearest historical analogue over the full

training set. In addition, k -nearest neighbours with $k = 5$ and logistic regression were trained on continuous log-return windows rather than bipolar inputs, because these methods do not require the Hopfield-specific binary representation. Standardization parameters were estimated from the training set only. Performance was evaluated primarily with balanced accuracy, which assigns equal importance to both return directions.

For classical Hopfield forecasting, each test context was processed in two ways. In direct retrieval, the most similar pattern was selected directly from the limited Hopfield memory. In Hopfield retrieval, the current context was first transformed by Hopfield dynamics and the nearest stored historical pattern was selected afterwards. In both cases, the actual historical continuation of the retrieved pattern was used as the forecast. Comparing the two variants isolates the contribution of Hopfield dynamics from the effect of merely restricting the historical memory.

Asymmetric temporal associative model

Classical Hopfield learning is symmetric and primarily encodes static pattern stability. Forecasting is directional, motivating an additional asymmetric temporal association experiment. Following the general idea of sequence association in asymmetric neural networks [12], consecutive training windows were paired as

$$X_t \rightarrow X_{t+1}. \quad (10)$$

The transition matrix was defined as

$$W_{\text{temp}} = \sum_t X_{t+1} X_t^T. \quad (11)$$

For a test context, one matrix update produced a predicted next state, and the final element of this state was interpreted as the predicted next return sign. This model is reported separately from the classical energy-minimizing Hopfield network because the asymmetric matrix does not possess the same classical Lyapunov-energy interpretation.

Finally, predictions from three representative scales, $L \in \{10, 50, 200\}$, were compared at aligned test dates. Cases in which all three temporal models predicted the same sign were labelled scale-agreeing. This test was used to examine whether cross-scale agreement was associated with more reliable forecasts; it was not intended as a formal test of financial multifractality.

Results

Descriptive properties of the financial series

The three assets exhibited substantially different return variability. TSLA consistently showed the largest dispersion of daily returns, SPY the smallest, and AAPL an intermediate level. Positive-return ratios were generally close to 0.5, so the sign representation did not exhibit an extreme permanent class imbalance. Yearly Hurst estimates varied across both assets and years, with values on both sides of 0.5. The lack of a stable Hurst regime reinforces the decision to treat temporal scale as an experimental factor rather than to assume a single persistent long-memory mechanism. The corresponding annual characteristics are shown in Fig. 1.

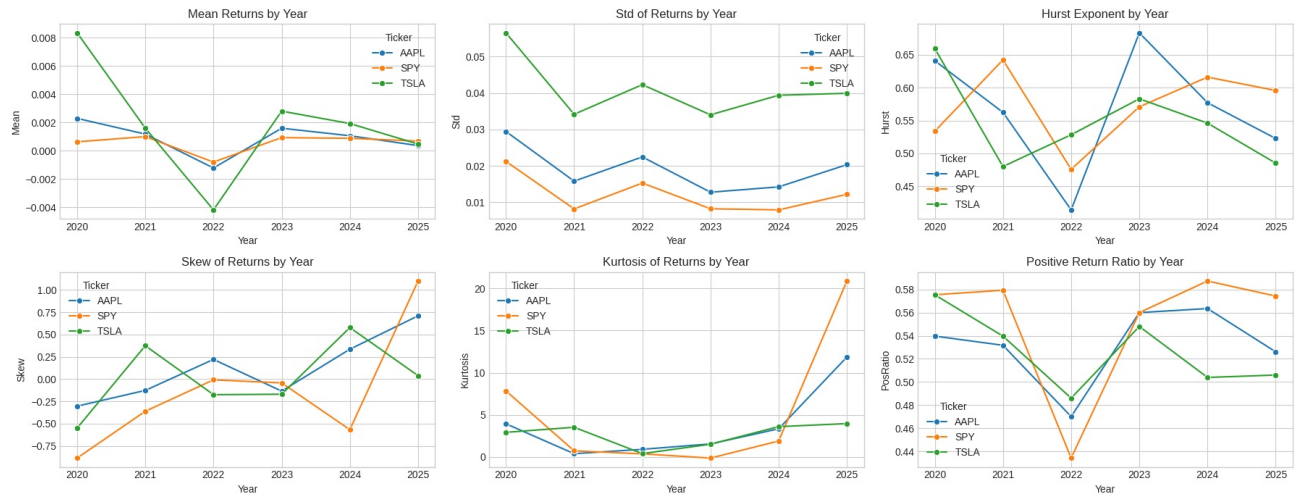


Figure 1. Descriptive characteristics of the investigated financial series. Annual mean log return, return standard deviation, Hurst exponent, skewness, kurtosis, and positive-return ratio for SPY, AAPL, and TSLA. The assets differ substantially in return variability, while the Hurst estimates vary across both assets and years rather than indicating a single stable long-memory regime.

Stored memories are recalled, but capacity matters

The synthetic validation recovered the corrupted reference pattern exactly and reduced the network energy, confirming the basic implementation. Financial stored-pattern experiments produced the expected qualitative behaviour: low-load memories were recalled reliably after moderate corruption, whereas increasing the number of stored patterns reduced stability and recall quality. For example, low-load short-window configurations could restore corrupted stored states to perfect accuracy, while high-load long-window configurations showed substantial degradation. This behaviour is consistent with interference among correlated memories and with the finite-capacity character of the classical model [1]. The joint dependence on corruption level, window length, and memory load is illustrated in Fig. 2.

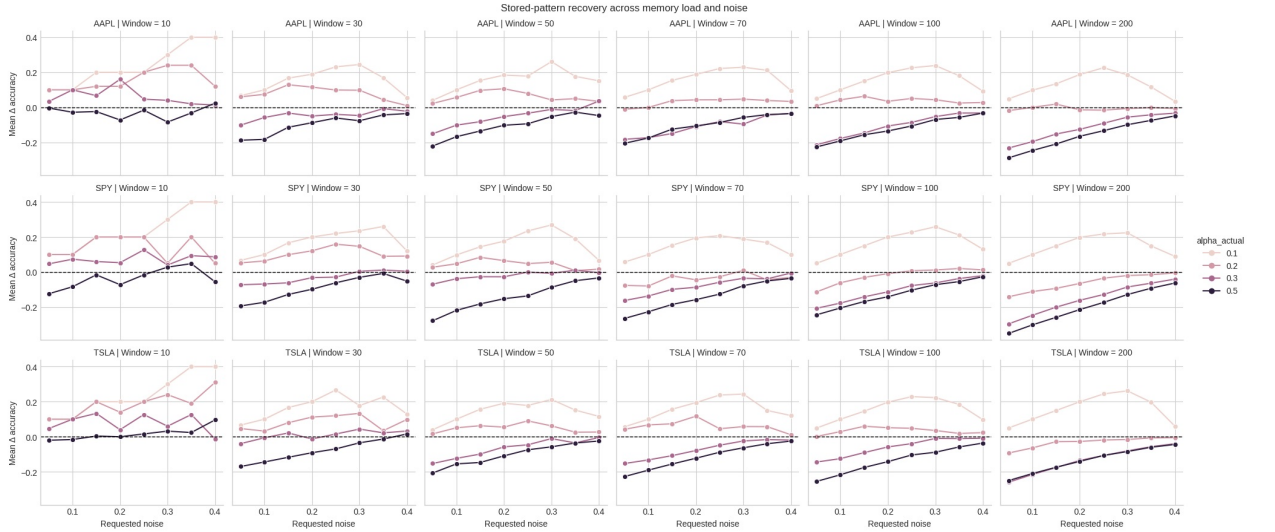


Figure 2. Stored-pattern recall under different memory loads and corruption levels. Recovery quality is shown across assets, temporal window lengths, and realized memory loads $\alpha = P/N$. Lower-load networks retain substantially stronger recall as corruption increases, whereas higher memory loads lead to increasing interference and degraded reconstruction.

The control experiment is important for interpreting later negative results. It demonstrates that failure on unseen test fragments is not explained by a network that is generally unable to store or retrieve the patterns on which it was trained.

Unseen financial windows are not reliably reconstructed

Before introducing artificial corruption, clean unseen test windows were passed through the trained network to determine whether they were stable under its dynamics. A test pattern was considered stable if a complete update sweep changed none of its bits. Most unseen financial windows were not preserved exactly: the network altered them even when no artificial noise had been introduced. Figure 3 shows how strongly the resulting states retained the original unseen patterns across memory loads and window lengths.

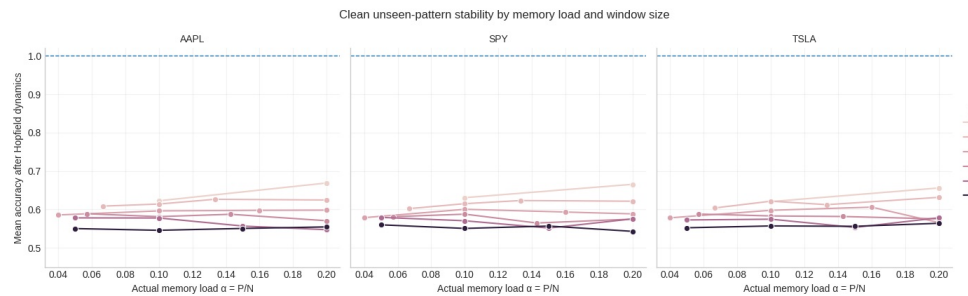


Figure 3. Behaviour of clean unseen patterns under classical Hopfield dynamics. Mean retained accuracy of previously unseen test patterns is shown as a function of memory load for different temporal window lengths and assets. A pattern is a fixed point only if a complete network sweep leaves all of its bits unchanged; the observed retained accuracies therefore show that unseen financial windows are generally modified by the trained network.

The clean-unseen experiment shows that the network does not simply leave

previously unseen market states unchanged. However, this does not imply that the dynamics is random. The network frequently moves unseen patterns closer to states supported by its stored historical memory. This associative movement is shown in Fig. 4.

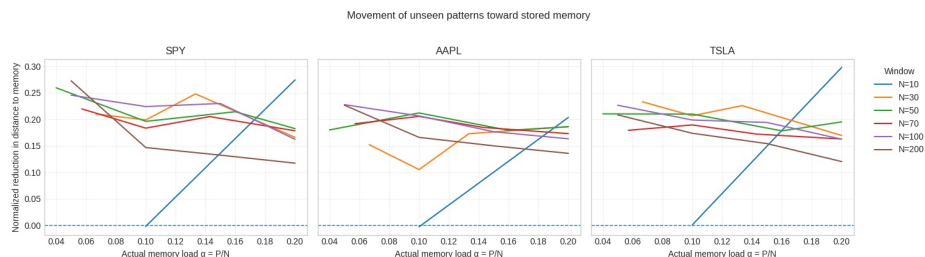


Figure 4. Movement of unseen patterns toward stored historical memory. The normalized reduction in Hamming distance to the nearest stored pattern after Hopfield dynamics is shown as a function of actual memory load. Positive values indicate movement toward the stored historical memory. Although unseen states are generally not stable themselves, Hopfield dynamics often moves them closer to memory-supported states.

The principal reconstruction experiment then tested whether this associative movement is useful when an unseen test fragment is artificially corrupted. Across the investigated configurations, the reconstruction metric ΔA was predominantly negative. The network therefore tended to alter the noisy input without recovering the original clean test state. All runs in the main noisy-unseen experiment converged under the chosen sweep limit, ruling out premature termination as the explanation for the negative result. Figure 5 summarizes the reconstruction outcome for the sign-only representation. The combined sign-volatility representation produced the same qualitative result (Supplementary Fig. S1).

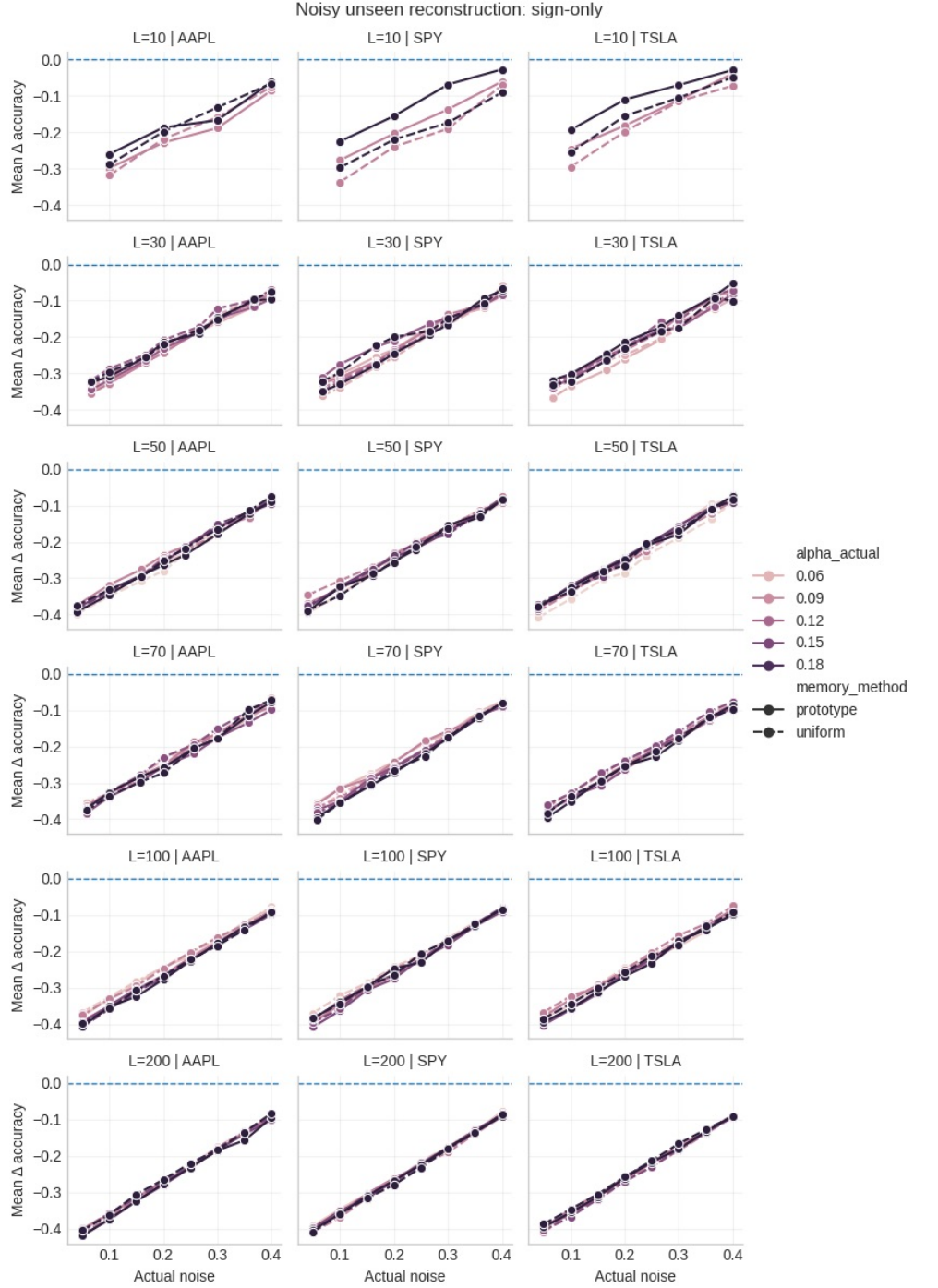


Figure 5. Reconstruction of noisy unseen sign patterns. Change in bitwise accuracy after classical Hopfield dynamics, $\Delta A = A(X_{\text{restored}}, X) - A(X_{\text{noisy}}, X)$, is shown against the realized corruption level across assets, window lengths, and memory loads. Values below zero indicate that the final network state is farther from the original clean unseen pattern than the corrupted input was. The predominantly negative values show that classical Hopfield dynamics does not systematically correct unseen financial fragments.

These results can be interpreted through a coverage–interference trade-off. At low memory load, stored patterns remain comparatively stable, but the memory represents only a small portion of historical diversity. Increasing the number of stored patterns improves historical coverage but also increases interference among stored states. Historical-analogue diagnostics support this interpretation: for short combined sign–volatility windows, the complete training set can contain a relatively close analogue of an unseen state even when the limited Hopfield memory does not contain that analogue. For long sign-only windows, even the closest state in the complete training history can remain substantially different from the test pattern.

Prototype-based memory selection partially improved historical coverage. In the combined sign–volatility representation, prototype selection reduced the mean distance to the nearest stored historical example by roughly 0.7–0.9 percentage points across the investigated loads (Supplementary Fig. S2). However, this improvement in memory composition did not translate into systematic successful reconstruction. The central reconstruction hypothesis is therefore not supported for the classical binary model in the investigated setting.

One-step directional forecasting remains weak

Historical similarity did not produce a stable forecasting advantage. The full-training nearest-analogue method occasionally outperformed simple rules for individual assets and window lengths, but the effect changed sign across configurations. Likewise, k -nearest neighbours and logistic regression operating on continuous return windows produced balanced accuracies largely around the 0.5 reference level, with isolated stronger and weaker cases. These results are consistent with the broader financial-forecasting literature, where reported machine-learning gains depend strongly on features, horizon, market regime, and validation design rather than appearing as a universal property of a model family [4, 9, 11]. The overall comparison is summarized in Fig. 6.

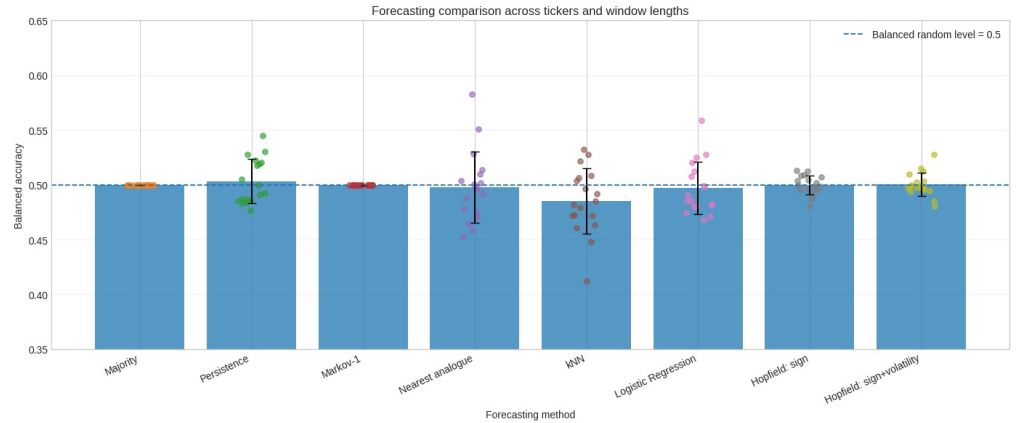


Figure 6. Comparison of one-step directional forecasting methods. Balanced accuracy is compared across simple baselines, historical-analogue retrieval, k -nearest neighbours, logistic regression, and classical Hopfield-based forecasting. Bars show the mean across the investigated asset/window configurations, while individual points represent configuration-level results. The horizontal reference at 0.5 denotes absence of balanced discrimination between the two return directions.

Classical Hopfield processing did not systematically improve direct retrieval from the same limited memory. Across sign-only and sign–volatility representations, memory

loads, and memory-selection strategies, positive and negative changes were both observed, while the median contribution of Hopfield dynamics remained close to zero. This absence of a systematic advantage is visible at the individual-configuration level for the sign-only representation in Fig. 7, with the corresponding combined-representation result provided in Supplementary Fig. S3; the aggregate results are summarized numerically in Table 1. Moving a current context toward an attractor supported by historical memory therefore does not imply that the associated historical continuation is informative about the next daily return sign.

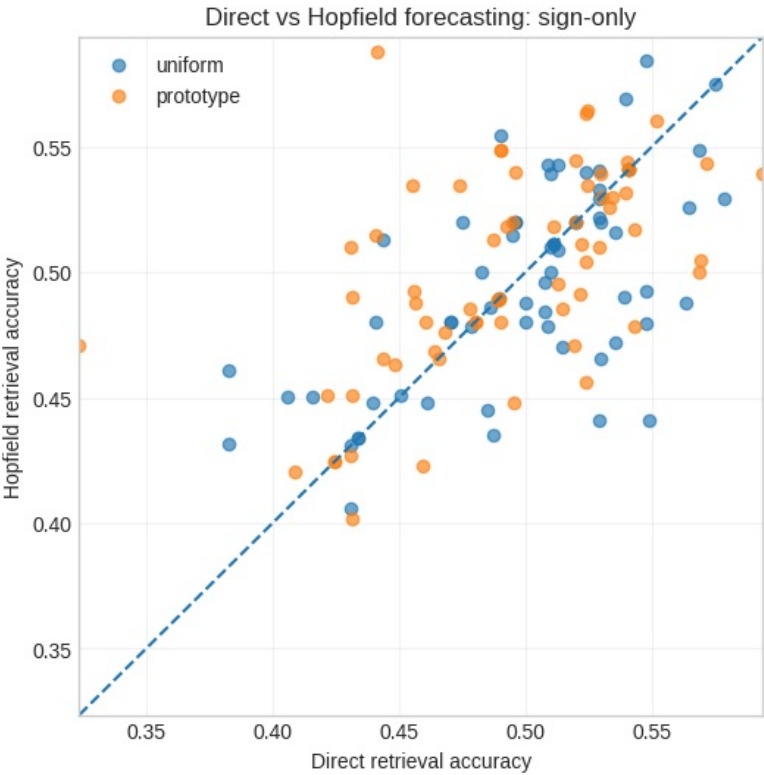


Figure 7. Direct historical retrieval versus Hopfield-assisted retrieval. Each point represents one experimental configuration. The horizontal coordinate is forecasting accuracy obtained by direct retrieval from the limited historical memory, and the vertical coordinate is accuracy after the same query was first transformed by classical Hopfield dynamics. The diagonal indicates equal performance. Points on both sides of the diagonal show that Hopfield dynamics does not provide a systematic forecasting advantage.

Table 1. Contribution of classical Hopfield dynamics to historical-analogue forecasting. Results are averaged across the investigated ticker, window-length, and memory-load configurations. Δ denotes Hopfield retrieval accuracy minus direct retrieval accuracy.

Representation	Memory	Direct	Hopfield	Mean Δ	Median Δ
Sign + volatility	Prototype	0.5025	0.5042	0.0017	0.0043
Sign + volatility	Uniform	0.4962	0.5061	0.0099	0.0000
Sign only	Prototype	0.4918	0.4999	0.0081	0.0000
Sign only	Uniform	0.5006	0.4964	-0.0042	0.0000

Temporal association produces asset- and scale-specific effects

The asymmetric temporal model produced more heterogeneous results than the classical static-memory forecast. For AAPL, balanced accuracy remained below 0.5 for all investigated window lengths, ranging approximately from 0.45 to 0.49. SPY achieved its strongest values at $L = 50$ and $L = 200$, with balanced accuracies of approximately 0.55 and 0.54, respectively. TSLA exceeded 0.5 for most investigated windows and reached approximately 0.59 at $L = 200$ and 0.55 at $L = 70$. The complete dependence on window length is shown in Fig. 8.

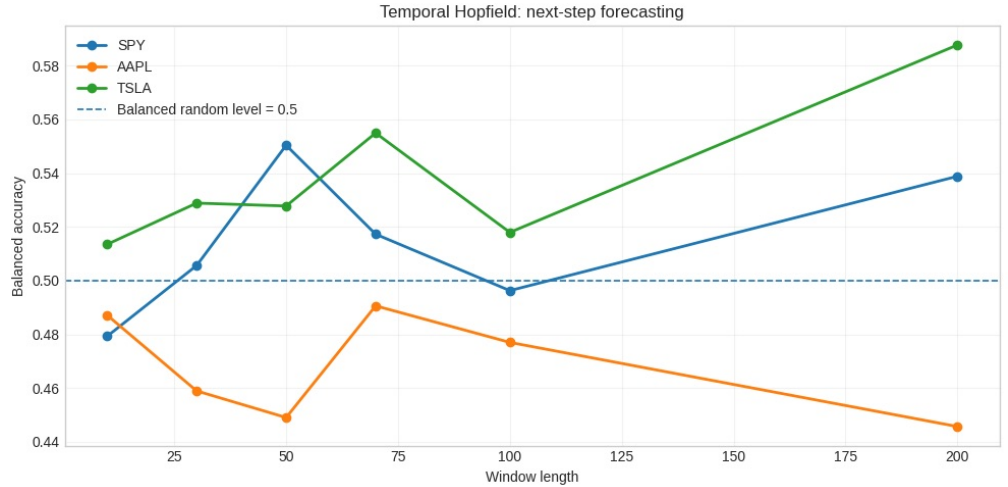


Figure 8. One-step forecasting with the asymmetric temporal associative model. Balanced accuracy is shown as a function of temporal window length for AAPL, SPY, and TSLA. The dashed reference line denotes 0.5 balanced accuracy. The effect of temporal context is asset-dependent: local improvements occur for SPY and TSLA, but no window length produces a consistent advantage across all three assets.

These isolated improvements should not be interpreted as evidence of a universally predictive temporal Hopfield model. The same architecture behaves differently across assets, and no monotonic relationship between window length and forecasting quality is present. Larger temporal context is therefore not sufficient by itself to improve directional forecasting.

Cross-scale agreement also failed as a confidence signal. For all three assets, agreement among the $L = 10$, 50, and 200 temporal models did not improve balanced accuracy relative to cases in which the scale-specific predictions disagreed (Table 2). Thus, agreement among the three investigated scales did not identify more reliable forecasts.

Table 2. Forecasting performance under cross-scale agreement. Predictions at $L = 10, 50$, and 200 were aligned to the same test dates. “Agreement” indicates that all three temporal models predicted the same sign.

Asset	Agreement	N	Accuracy	Balanced accuracy
AAPL	No	61	0.459	0.457
AAPL	Yes	41	0.390	0.376
SPY	No	55	0.545	0.543
SPY	Yes	47	0.596	0.518
TSLA	No	60	0.650	0.650
TSLA	Yes	42	0.476	0.457

Discussion

The experiments reveal a sharp distinction between associative recall and generalization. The classical Hopfield network performs its native task successfully in the low-load regime: stored financial patterns can be stable attractors and can be recovered after corruption. The difficulty appears when this mechanism is expected to reconstruct a new financial state that was not itself stored.

This distinction is theoretically important. The classical model encodes a finite set of attractors, not a supervised mapping from corrupted examples to arbitrary unseen targets. Generalization beyond stored examples can occur under additional structural assumptions, as demonstrated in random-feature Hopfield models where training and test examples are generated by shared latent features [3]. The present financial windows do not provide evidence that an analogous generalization regime emerges naturally from simple sign-based Hebbian storage.

The coverage–interference trade-off provides a concrete interpretation. A small stable memory cannot represent the diversity of historical financial contexts, whereas enlarging the memory eventually destabilizes the stored patterns. Prototype selection addresses only the first side of this trade-off. It yields slightly more representative memories, particularly for the combined sign–volatility representation, but does not alter the fundamental interference mechanism of the classical weight matrix.

The representation itself is another limitation. Sign-only encoding intentionally discards return magnitude, and the binary volatility extension retains only a coarse magnitude category. This simplification is appropriate for the classical bipolar Hopfield state space but can remove information available to conventional forecasting methods. The weak results of logistic regression and k NN on continuous returns suggest, however, that representation loss is not the only explanation for weak next-day directional performance in this dataset.

The temporal experiment indicates that directionality matters. A symmetric classical Hopfield network is designed to relax toward static attractors, whereas a forecasting problem asks for a transition from the present context to a future state. Asymmetric sequence-association models were developed precisely to encode such transitions [12]. The observed asset-specific gains for SPY and TSLA show that this modification can change the forecasting behaviour substantially, although the lack of cross-asset consistency prevents a stronger conclusion.

The role of temporal scale is similarly nuanced. Financial data have long motivated scale-dependent and multifractal descriptions [7, 8], but scale dependence should not be equated with exploitable directional predictability. In this study, changing L affected both reconstruction and forecasting, yet no universal best scale emerged, and cross-scale forecast agreement did not improve reliability. The results therefore support only the

narrower statement that model behaviour depends on temporal scale; they do not constitute a test or confirmation of a multifractal market model.

Limitations

The study uses only three liquid US assets and daily data from a single six-year period. Consequently, the results should not be interpreted as universal statements about all markets, frequencies, or regimes. The test set is a single chronological holdout rather than a full walk-forward evaluation across multiple market regimes.

The principal Hopfield representations are strongly compressed: sign-only patterns remove return magnitude, while the volatility channel is binary. The prototype-memory experiment uses clustering as a pragmatic representative-sample strategy rather than a formal financial motif-discovery method. Comparisons between sign-only and sign-volatility Hopfield networks are also not strict causal representation ablations, because concatenating the volatility channel changes the number of neurons and, at fixed load α , changes the absolute number of stored patterns.

Finally, the temporal asymmetric experiment is intentionally simple and should not be conflated with modern sequence models or with the original symmetric Hopfield energy dynamics. Its purpose is to test whether directional association changes the empirical conclusion, not to establish a state-of-the-art forecasting architecture.

Future Work

Several extensions follow directly from the limitations identified in this study. First, continuous-valued and modern Hopfield formulations should be investigated to avoid the strong information compression imposed by bipolar sign and volatility representations and to increase effective associative-memory capacity. Modern Hopfield architectures with learned representations are particularly relevant because they can combine associative retrieval with high-dimensional continuous states.

Second, the temporal evaluation should be extended from a single chronological holdout to walk-forward testing across multiple market regimes, assets, and sampling frequencies. This would allow the stability of the observed asset- and scale-specific effects to be assessed more rigorously.

Third, memory formation can be improved beyond uniform sampling and simple k -means prototypes. Financial motif discovery or learned representative states could provide historical memories that better cover recurring market regimes while limiting redundant and strongly correlated patterns.

Finally, the multi-scale analysis can be extended beyond agreement among independently trained window lengths. A model that jointly combines short-, medium-, and long-scale representations may provide a more direct test of whether scale-dependent financial structure can be exploited by associative memory.

Conclusion

This study evaluated whether a classical binary Hopfield network can exploit recurring local structures in financial time series for reconstruction and one-step direction forecasting. The implementation successfully stored and recovered trained financial patterns at low memory load, confirming that the basic associative-memory mechanism operates as expected. However, this success did not generalize to previously unseen noisy financial fragments: Hopfield dynamics typically moved test patterns toward memory-supported attractors without reconstructing the original clean state.

The main limitation is a trade-off between historical coverage and memory interference. Small memories remain stable but represent too little of the historical pattern space, while larger memories improve coverage at the cost of attractor degradation. Alternative binary representations and prototype-based memory selection modify this trade-off but do not remove it.

For next-day directional forecasting, neither classical Hopfield retrieval nor the conventional baselines demonstrated a stable cross-asset advantage. A simple asymmetric temporal associative model produced stronger results for selected SPY and TSLA windows, but the effect remained asset- and scale-dependent. No single temporal scale was consistently superior, and agreement across short, medium, and long windows did not identify more reliable forecasts.

Overall, the results suggest that classical Hopfield networks are effective as associative memories for stored financial patterns but are limited as direct generalizers to new market states. These results establish a classical associative-memory baseline against which more expressive Hopfield formulations and temporally structured models can be evaluated.

Data and Code Availability

The complete implementation and experiment notebook are publicly available in the project GitHub repository:

<https://github.com/alyonaprikhodko/hopfield-financial-time-series/>. The experiments use publicly available historical market data obtained through the Yahoo Finance interface. The repository contains the code required to reproduce the data preprocessing, reconstruction, forecasting, and evaluation experiments reported in this study.

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Supporting Information

Supplementary reconstruction results

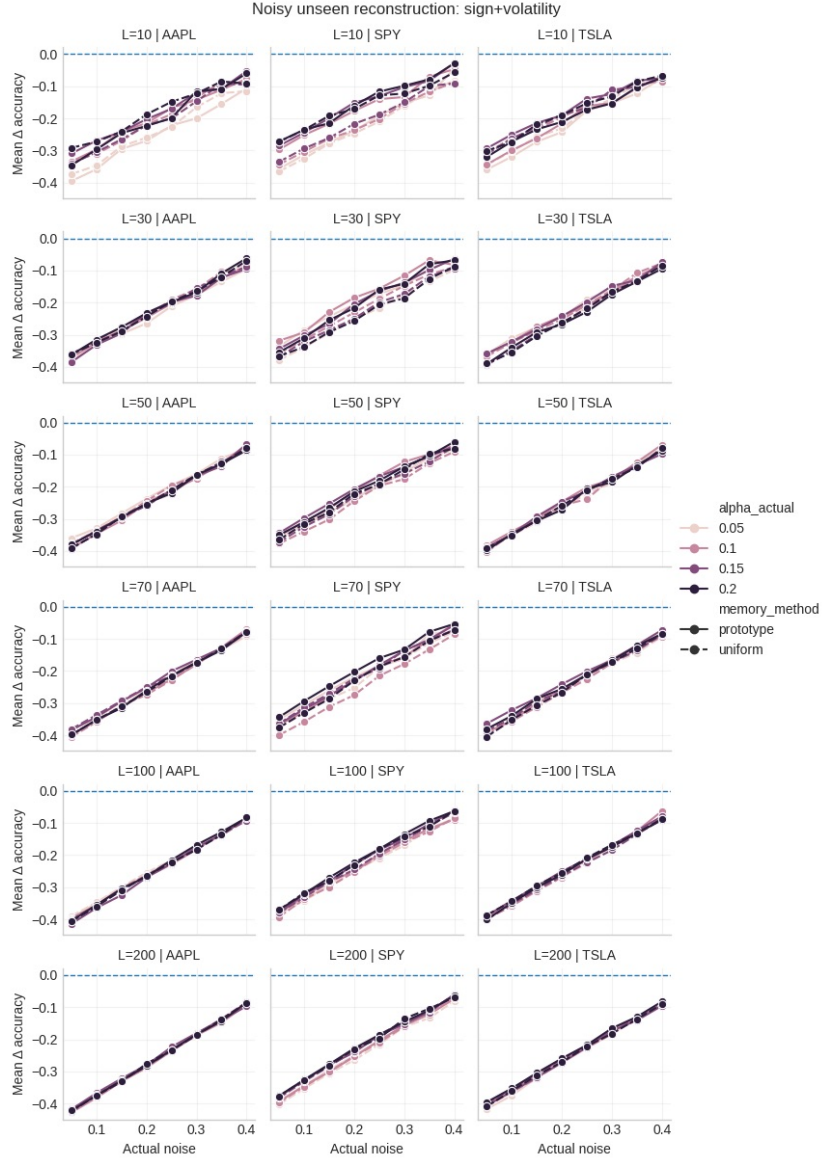


Figure S1. Reconstruction of noisy unseen patterns using the combined sign–volatility representation. Change in reconstruction accuracy after classical Hopfield dynamics across assets, window lengths, memory loads, and realized corruption levels. As in the sign-only experiment, the predominantly negative changes indicate that Hopfield dynamics does not systematically recover the original clean unseen state.

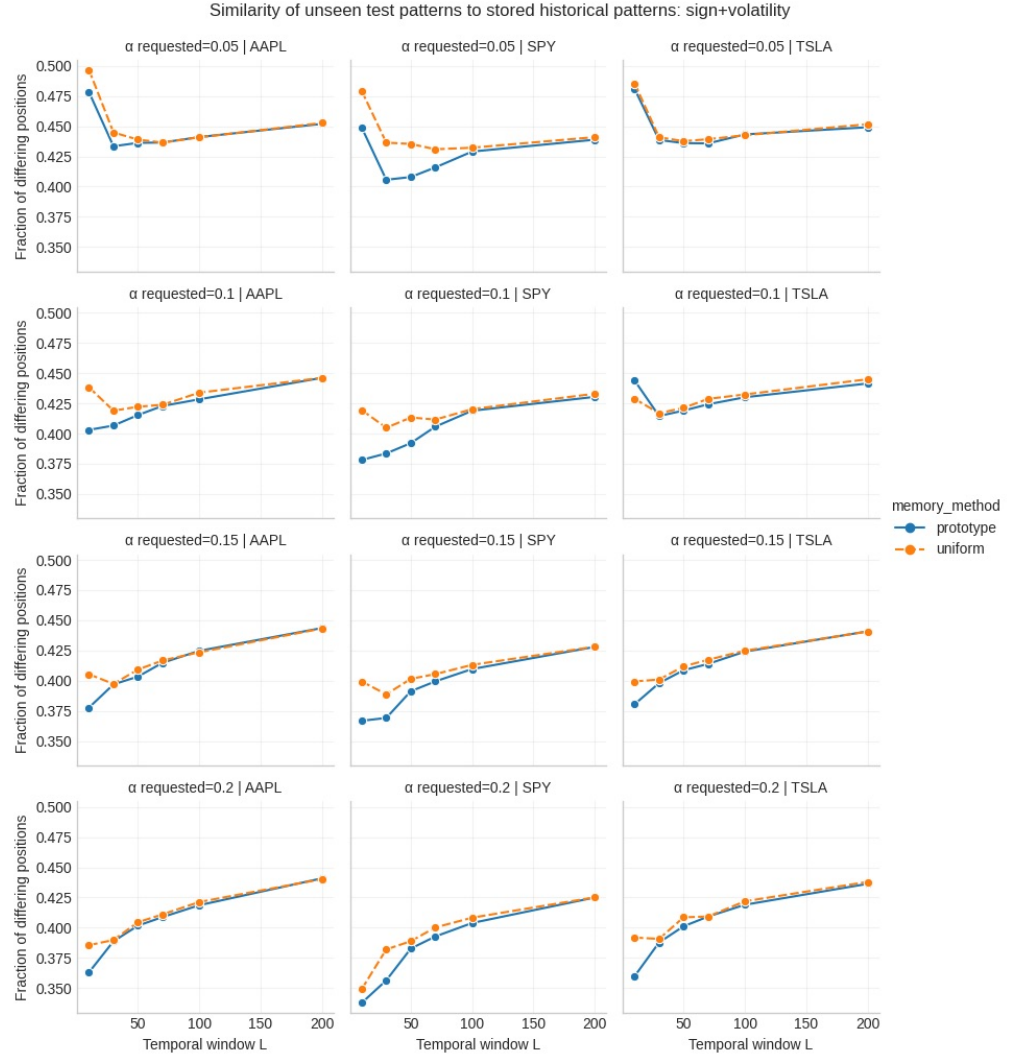


Figure S2. Historical coverage of unseen patterns by uniform and prototype memories. The mean normalized Hamming distance from each clean unseen test pattern to its closest stored historical pattern is shown for the combined sign-volatility representation. Lower values indicate better historical coverage. Prototype-based memory selection generally reduces this distance at low and moderate memory loads, although the improvement does not lead to successful unseen-pattern reconstruction.

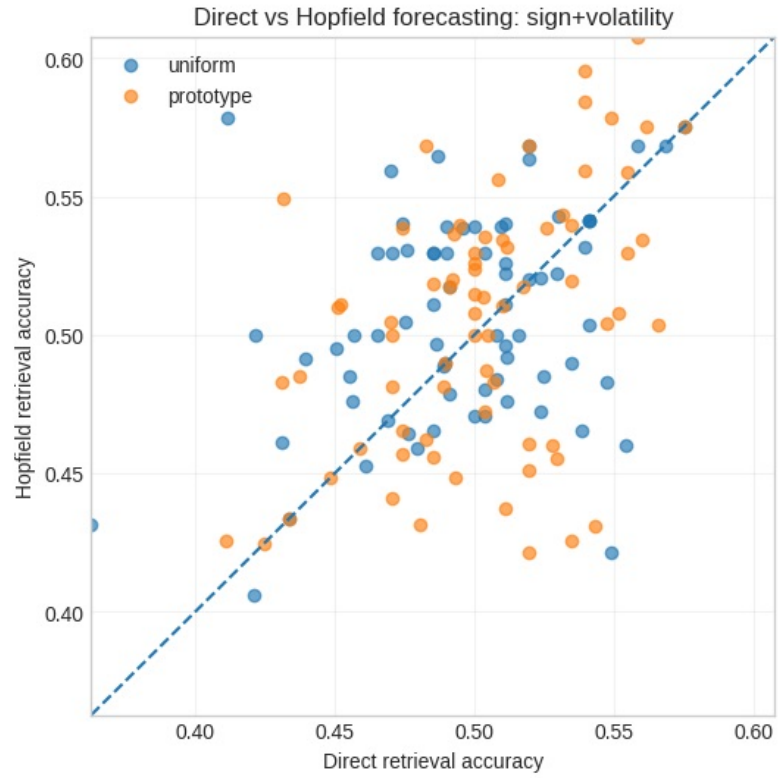


Figure S3. Direct historical retrieval versus Hopfield-assisted retrieval for the combined sign–volatility representation. Each point represents one experimental configuration. Points occur on both sides of the equality line, confirming that Hopfield dynamics does not provide a systematic forecasting advantage for the combined representation.